Tradelink Functional Specification

Field	Value
Document	Functional Specification Document (FSD)
Product	Tradelink, with Tradelink AI
Version	1.0
Date	4 November 2026
Status	Draft for review
Owner	Janya Mali
Related documents	User Guide, Technical Specification, User stories, Decision records

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1. Document control

1.1 Purpose

This document defines what Tradelink does today and the business rules behind it. It is an as-built specification: every statement is based on the code, the database migrations, the tests or an accepted decision record. Features that were planned but not built are marked **Not built** and listed in section 18.

1.2 Audience

Reader	Uses this document to
Product owner	Confirm scope, rules and open decisions
Design	Check screens, states and copy against behaviour
QA	Derive test cases from requirements and acceptance criteria
Business reviewers and interviewers	Understand the product and the reasons behind it

Engineers should read the Technical Specification for how each part is built.

1.3 Scope of this document

Covered	Not covered
The buyer, supplier and admin web app (`/buyer`, `/supplier`, `/admin`)	Code structure, database design and deployment (see the Technical Specification)
Sign-in, sign-up and password reset	Step-by-step help for users (see the User Guide)
Tradelink AI, the assistant inside the app	The separate Tradelink AI shop foundation at `/r`, which the main app does not use. It is described only in the Technical Specification.

1.4 Sources

Source	Used for
`app/` routes and `src/` code	Screens, actions, rules and copy
`supabase/migrations/` (17 applied migrations)	Permissions, state models, business rules and notifications
Unit tests (`src/**/*.test.ts`), e2e tests (`e2e/`) and the concurrency test (`tests/concurrency/`)	Test references
`docs/USER_STORIES.md`	User story IDs
`docs/adr/`	Decision reasons

Where the code and older documents disagree, this document follows the code. The differences are listed in the Technical Specification appendix "Differences from older documents".

1.5 ID conventions

Prefix	Meaning
F-xx	Feature
FR-xxx	Functional requirement
BR-xxx	Business rule
NFR-xxx	Non-functional requirement
SCR-xx	Screen
J-xx	User journey
DEC-xx	Product decision
SH, AUTH, BUY, SUP, ADM, AI, OPS	User story IDs from `docs/USER_STORIES.md`

1.6 Document history

Version	Date	Author	Change
1.0	4 November 2026	Janya Mali (with Claude)	First as-built specification, from an audit of the code at commit 9250e98

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2. Product overview

2.1 Problem statement

Small food businesses in Singapore order stock from wholesale suppliers over WhatsApp and phone calls. Orders get lost in chat, nobody knows what stock is actually available, short or damaged deliveries are hard to follow up, and spending is hard to track. Suppliers re-type orders by hand and find out about stock problems late.

2.2 Goals and non-goals

Goals	Non-goals
One place where buyers order from all their suppliers and suppliers manage those orders	Paying suppliers or collecting payment (no payment features at all)
Stock that is checked and reserved when an order is placed, so suppliers are not oversold	Accounting, tax filing or invoicing
A clear record of each delivery, each problem and each credit	Delivery routing or driver tracking
An assistant (Tradelink AI) that does what the person could do in the app, but only after they confirm	AI that sends orders or changes data on its own
A platform team that sees what needs attention and can resolve disputes	A public marketplace or supplier discovery

2.3 Value by user type

User	What Tradelink gives them today
Buyer (restaurant, café, bakery)	Order by typing to Tradelink AI or from usual items. One review step with stock, minimums, cut-offs and budget. Delivery checks, problem reports and credit tracking. Weekly spend insights.
Supplier (wholesaler)	Orders arrive with stock already reserved. Accept, part accept (now or as a second delivery) or reject. Inventory with a full stock history. Problems to answer in one list.
Admin (Tradelink operations)	A work queue of what needs attention, assignment to operators, dispute resolution, supplier suspension, and an append-only audit log.

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3. Scope

Area	In scope	Out of scope	Reason
Accounts	Email and password sign-in, sign-out, password reset, self sign-up for buyers and suppliers	Two-factor sign-in, invitation-only onboarding, team members	Owner decision to make Tradelink easy to try (ADR 0024)
Ordering	Tradelink AI, usual items, product browsing, cart review, one order per supplier, stock reservation	Minimum-order negotiation and delivery fees (BUY-25, SUP-08)	Not built. The design is in ADR 0020.
Fulfilment	Accept, part accept, reject, out for delivery, delivered, a second delivery for the rest, delivery date changes, order messages	Delivery routing	Not part of the product brief
After delivery	All here, Not here yet, problem reports, supplier answers, credits, disputes	Invoices: upload, reading, checks, bank-detail warnings, manual entry	Not built. An `invoice` problem type exists in the database, but no flow creates it.
Payments	None	Paying invoices, subscriptions, Stripe	The product never moves money. The owner decided on 4 November 2026 not to integrate Stripe at all.
Suppliers not on Tradelink	A buyer can add one through Tradelink AI. It is saved as an *invited* business, and admins see a follow-up after 24 hours.	Invitation emails, accept or decline pages, claiming the invited record	Not built. ADR 0024 replaced invitations with self sign-up "until invitations are built".
Notifications	In-app notifications, the bell panel and the Alerts page for every role	Email, SMS and web push	Push code exists but nothing turns it on or sends it. There is no email provider.
Admin	Needs attention (with lead and operator visibility), disputes, supplier suspend and reactivate, read-only orders, users, buyers and operators, audit log	Approving suppliers, inviting buyers, managing users or operators, system status page, connecting a buyer and a supplier	Not built

Area	In scope	Out of scope	Reason
AI	Tradelink AI chat for buyers and suppliers, with read tools and confirm-before-write tools	Invoice reading, negotiation drafts, an evaluation set	Not built
Devices	Responsive web app for phone and laptop, an offline page, an app manifest	Native apps, an install prompt, offline ordering	See DEC-01

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4. Users and permissions

4.1 Roles

Role	Description	Main goals
Buyer	A member of a buyer business (organisation kind `buyer`). One person belongs to one business.	Keep the kitchen stocked, order quickly, check deliveries, get credit for problems, stay within budget
Supplier	A member of a supplier business (kind `supplier`)	Turn orders into deliveries, keep stock and prices right, answer problems
Admin: operations lead	A Tradelink operator whose `admin_profiles.is_lead` flag is set (Hannah Lim in the demo data)	See the whole attention queue, assign items, resolve disputes, oversee suppliers
Admin: operator	Any other Tradelink operator (kind `platform`)	Work through the attention items assigned to them
Not signed in	Anyone	Sign in, create an account, reset a password

The role comes from the business the person belongs to. There is no per-person role inside a business (owner or staff) in the main app.

4.2 Permissions matrix

Y = allowed, N = refused, Own = only their own records, Conn = only with businesses they trade with.

Action	Buyer	Supplier	Admin lead	Admin operator	Enforced by
Sign in, sign out, reset password	Y	Y	Y	Y	Supabase Auth
Create an account by sign-up	Y (as a buyer business)	Y (as a supplier business)	N (admin accounts cannot be created by sign-up)	N	`app.handle_new_user` trigger
Open another role's pages	N (redirected, logged as denied)	N	N	N	Request proxy and `requireRole`

Action	Buyer	Supplier	Admin lead	Admin operator	Enforced by
See products and prices	Conn	Own	Y	Y	RLS `can_see_product`
Use Tradelink AI	Y	Y	N	N	`/api/ai/chat` route check
Save a cart	Own	N	N	N	`save_draft`
Place orders	Y (Conn suppliers)	N	N	N	`place_orders`
Cancel an order	Own, while pending	N	N	N	`cancel_order`
Accept, part accept or reject an order	N	Own orders	N	N	`accept_order`, `part_accept_order`, `reject_order`
Mark out for delivery or delivered	N	Own orders (also while suspended)	N	N	`mark_out_for_delivery`, `mark_delivered`
Send or deliver the rest	N	Own orders (also while suspended)	N	N	`mark_rest_out_for_delivery`, `deliver_rest`
Cancel the rest	Own orders	Own orders	N	N	`cancel_rest`
Ask to move a delivery date	Own orders	Own orders	N	N	`propose_delivery_date`
Answer a date request	The side that did not ask	The side that did not ask	N	N	`answer_delivery_date`
Send an order message	Own orders	Own orders	N	N	`send_order_message`
Confirm All here	Own delivered orders	N	N	N	`confirm_all_here`
Report Not here yet	Own orders out for delivery	N	N	N	`report_not_here`
Report a problem	Own delivered orders	N	N	N	`report_problem`
Answer a problem	N	Own	N	N	`answer_issue`
Mark a credit received	Own	N	N	N	`mark_credit_received`
Resolve a dispute	N	N	Y	Y	`resolve_dispute`
Add a product for a supplier	Conn (active or invited supplier)	N	N	N	`buyer_add_product`
Add a supplier not on Tradelink	Y	N	N	N	`buyer_add_supplier`

Action	Buyer	Supplier	Admin lead	Admin operator	Enforced by
Add a product to own catalogue	N	Own	N	N	`supplier_add_product`
Receive stock, confirm a buyer-added product, change price, adjust stock	N	Own	N	N	`receive_stock`, `confirm_product_stock`, `change_price`, `adjust_stock`
Edit business profile and terms	N	Own	N	N	`update_business_profile`
Edit budget, alert switches and usual items	Own	N	N	N	`update_buyer_settings`, `set_usual_item`, `remove_usual_item`
Acknowledge a price change	Own	N	N	N	`ack_price_change`
See own notifications and mark read	Y	Y	Y	Y	RLS, `mark_notifications_read`
See the attention queue	N	N	All items	Only items assigned to them	`v_attention` view
Assign an attention item	N	N	Y	N	`assign_attention`
Mark an attention item handled	N	N	Y	Only items assigned to them	`mark_attention_handled`
Suspend or reactivate a supplier	N	N	Y	Y	`set_supplier_status`
See all orders, suppliers, buyers, users and operators	N	N	Y	Y (operator workload counts are shown to the lead only)	RLS `is_admin`, page code
Read the audit log	Rows about their business (via the database)	Own activity page	Y	Y	RLS on `audit_log`
Read stock history	N	Own products	Y	Y	RLS on `stock_movements`
Change theme	Y	Y	Y	Y	Browser cookie

Note: the database lets a buyer read audit rows that involve their business. No buyer screen shows them.

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5. Feature catalogue

Priority comes from the user story (Must or Should). Status: **Built**, **Partial** (built with gaps listed) or **Not built**.

ID	Feature	Role	Priority	Status	Related user stories
F-01	Sign in and sign out	All	Must	Built	AUTH-01, AUTH-02
F-02	Reset a forgotten password	All	Must	Built	AUTH-03
F-03	Create an account (self sign-up)	Buyer, supplier	Must	Built	AUTH-10
F-04	Role-based access and routing	All	Must	Built	AUTH-06, AUTH-09
F-05	Two-factor sign-in	All	Must, Should	Not built (removed by ADR 0024)	AUTH-04, AUTH-05, AUTH-08
F-06	Accept or decline an invitation	Supplier, buyer	Must	Not built	AUTH-07, SUP-20, SUP-21, ADM-15
F-07	App shell, navigation and responsive layout	All	Must	Built	SH-01, SH-04, SH-12
F-08	Home with Next up for every role	All	Must	Built	SH-02, BUY-01, BUY-02, BUY-03, BUY-04, BUY-07, BUY-08, BUY-09, SUP-01, SUP-02, ADM-01
F-09	Light, dark and system theme	All	Must	Built (remembered per browser, not per account)	SH-03
F-10	In-app notifications, bell panel and Alerts page	All	Must	Built	SH-09, ADM-08
F-11	Web push notifications	All	Should	Not built (code exists, nothing sends)	SH-09, SH-13
F-12	Offline page and app manifest	All	Should	Partial (no install prompt, no offline actions in the main app)	SH-07, SH-13
F-13	Loading, empty and error states	All	Must	Built	SH-05, SH-06
F-14	Audit log	Admin	Must	Built	SH-11, ADM-18
F-15	Tradelink AI assistant	Buyer, supplier	Must	Built	BUY-45, BUY-47, BUY-48, BUY-49, BUY-50, AI-04, AI-05
F-16	Order from typed text	Buyer	Must	Partial (through Tradelink AI tools; the rule-based fallback reader is not reachable)	BUY-10, BUY-11, AI-01, AI-02, AI-03

ID	Feature	Role	Priority	Status	Related user stories
F-17	Usual items and "Place tomorrow's order"	Buyer	Must	Built	BUY-08, BUY-12, BUY-46
F-18	Browse products and add to cart	Buyer	Must	Built	BUY-13, BUY-42
F-19	Cart review and place orders	Buyer	Must	Built	BUY-14, BUY-15, BUY-16, BUY-17, BUY-20, BUY-22, BUY-23
F-20	Below-minimum warning	Buyer, supplier	Must	Built (warning only; negotiation not built)	BUY-24
F-21	Add a missing product	Buyer	Must	Built	BUY-18
F-22	Add a supplier not on Tradelink	Buyer, admin	Must	Partial (record and admin follow-up only; no invitation is sent)	BUY-19, ADM-06
F-23	Orders list and order page	Buyer	Must	Built (orders only; no invoices)	BUY-27
F-24	Cancel a pending order	Buyer	Must	Built	BUY-28
F-25	Delivery check: All here and Not here yet	Buyer	Must	Built	BUY-29, BUY-31
F-26	Report a problem	Buyer	Must	Built	BUY-30
F-27	Credits and confirming a credit arrived	Buyer, supplier	Must	Built	BUY-37
F-28	Invoices: upload, read, check, bank details, manual entry	Buyer, supplier	Must	Not built	BUY-05, BUY-06, BUY-26, BUY-32, BUY-33, BUY-34, BUY-35, BUY-36, SUP-19, AI-07
F-29	Insights	Buyer	Must, Should	Built	BUY-38, BUY-39, BUY-40, BUY-41
F-30	Buyer settings: budget, alerts and usual items	Buyer	Must	Built	BUY-43, BUY-44, BUY-46
F-31	Incoming orders: accept, part accept, reject	Supplier	Must	Built	SUP-03, SUP-04, SUP-05, SUP-06
F-32	Deliveries, including a second delivery for the rest	Supplier, buyer	Must	Built	SUP-07, BUY-21
F-33	Delivery date changes and order messages	Buyer, supplier	None (added after the stories)	Built	BUY-21
F-34	Answer problems	Supplier	Must	Built	SUP-09
F-35	Inventory: add product, receive, confirm, change price, adjust	Supplier	Must, Should	Built	SUP-10, SUP-11, SUP-12, SUP-13, SUP-14, SUP-15

ID	Feature	Role	Priority	Status	Related user stories
F-36	Stock history and activity	Supplier	Must	Built	SUP-16, SUP-17, OPS-01
F-37	Supplier business profile	Supplier	Must	Built	SUP-18
F-38	Needs attention, with lead and operator visibility	Admin	Must	Built	ADM-02, ADM-03, ADM-04, ADM-05
F-39	Resolve a dispute	Admin	Must	Built	ADM-07
F-40	All orders, suppliers, low stock, users, buyers, operators	Admin	Must	Built (view only, except suspend and reactivate)	ADM-09, ADM-10, ADM-11, ADM-12, ADM-13, ADM-14, ADM-16, SUP-22
F-41	Admin settings	Admin	Must	Partial (account and appearance only)	ADM-19
F-42	System status page	Admin	Must	Not built (a liveness endpoint only)	ADM-17, OPS-05
F-43	Minimum-order negotiation and fees	Buyer, supplier	Must	Not built	BUY-25, SUP-08, AI-08
F-44	Team members	Buyer, supplier	Should	Not built	SH-14
F-45	Live updates	All	Should	Not built in the main app	SH-10
F-46	Connect a buyer and a supplier (admin)	Admin	Should	Not built	ADM-20
F-47	No overselling, idempotent actions, state machines	System	Must	Built	OPS-02, OPS-03, OPS-04
F-48	Rate limits	System	Must	Partial (AI chat and order messages only)	OPS-07
F-49	Demo data and accounts	System	Must	Built	OPS-08
F-50	Email delivery, retention jobs, CI, AI evaluation set, performance budget	System	Must, Should	Not built	OPS-09, OPS-10, OPS-11, OPS-12, AI-06

6. Functional requirements

Test references point to real files. "No automated test" means the behaviour is only checked by hand or by reading the code.

6.1 Accounts and access

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-001	Sign in with email and password	All	Submit the sign-in form	The person lands on their role's home, or on the safe `next` path they came from	Wrong details show "Check your email and password."; too many tries show "Too many attempts. Try again in a few minutes."	`src/features/auth/routing.test.ts`, `e2e/smoke.spec.ts`
FR-002	Sign out	All	Sign out in the account menu or Settings > Account	The session ends and the sign-in page shows	Opening a role page afterwards leads to sign-in	No automated test
FR-003	Request a password reset	All	Submit an email on Forgot password	A reset email is requested from Supabase Auth; the page always says it was sent	The reply is the same whether the email exists or not	`src/features/auth/origin.test.ts`
FR-004	Set a new password	All	Open the emailed link and submit the form	The password changes and the person is signed in	A password under 12 characters is refused with "Use at least 12 characters."; an expired link shows an expired notice	`src/features/auth/password.test.ts`
FR-005	Create a buyer or supplier account	Not signed in	Submit Sign up (kind, business name, name, phone, email, password)	An active business, profile and membership are created; the business is connected to every active business of the other kind; admins get "New account"	A duplicate business name of the same kind is refused ("That business name may already be in use"); admin accounts cannot be created	`src/features/auth/signUp.test.ts`
FR-006	Send each person to their own area	All	Open any page	Buyers stay under `/buyer`, suppliers under `/supplier`, admins under `/admin`; others are redirected and the attempt is logged as denied	A buyer opening `/admin` lands on `/buyer`	`src/features/auth/routing.test.ts`
FR-007	Never redirect to another site after sign-in	All	Sign in with a `next` parameter	Only same-site paths are followed	`next=https://evil.example` lands on the role home	`src/features/auth/routing.test.ts`

6.2 Shared app behaviour

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-010	Show role navigation	All	Any signed-in page	Laptop (760px and wider): sections in the top bar. Phone: bottom tabs, with a More sheet when a role has more than 5 sections.	Buyer: Home, Tradelink AI, Orders, Products, Insights. Supplier: Home, Tradelink AI, Orders, Inventory, Activity. Admin: Home, Attention, Orders, Suppliers, Users, Operators, Audit log.	`src/features/shell/nav.test.ts`
FR-011	Show counts on navigation	All	Any signed-in page	Buyer Orders shows deliveries to check; supplier Orders shows new orders; supplier Inventory shows products to reorder; admin Attention shows visible items	Counts match the lists they lead to	`src/features/shell/nav.test.ts`
FR-012	Switch theme	All	Top-bar toggle or Settings > Appearance (System, Light, Dark)	The theme changes at once and is remembered in this browser	Reloading keeps the choice; System follows the device	No automated test
FR-013	Show in-app notifications	All	A database event for the person's business	A bell panel lists recent notifications with one action button each (for example Reply, Answer, Confirm credit, Check delivery)	The button opens the right order section (`#messages`, `#problems`)	`src/features/notifications/routes.test.ts`
FR-014	Mark notifications read	All	Open Alerts and choose Mark all read	All of the business's notifications are marked read	The unread count drops to 0	No automated test
FR-015	Show empty and error states	All	A list with no rows, or a page that fails	A plain message and a next step; failures show "This page did not load" with Try again	No page shows a blank area or a stack trace	`e2e/pages.spec.ts`
FR-016	Keep failed admin saves on the page	Admin	A save that cannot reach the server (for example a page opened before a deploy)	"This did not save. Reload the page and try again." appears under the control; the choice goes back to the saved value	The page stays usable	No automated test (checked by hand on 4 November 2026)
FR-017	Show an offline page	All	The network fails while loading a page	The service worker serves `/offline` with Try again	Works with no network	`e2e/smoke.spec.ts`

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-018	Fit phone screens	All	Any page at phone width	No sideways scrolling	Every buyer and supplier page passes the overflow check	`e2e/pages.spec.ts`

6.3 Buyer: ordering

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-020	Show what needs the buyer on Home	Buyer	Open Home	Next up shows one step at a time: tomorrow's order plan, deliveries to check, and to-dos (price changes, part deliveries, replies to problems, credits to confirm, date requests, messages)	With nothing to do: "Nothing needs you right now. Your usual items are covered and every delivery is checked."	`src/domain/todos.test.ts`, `src/domain/home.test.ts`
FR-021	Suggest tomorrow's order	Buyer	Usual items running out within 3 days	A plan tops each item up to cover 7 days, limited by available stock; buttons Place orders, Review in cart, Not today	Not today hides the plan for the day; changing "in your shop" brings it back	`src/domain/reorder.test.ts`, `src/features/buyer/usual-days.test.ts`
FR-022	Order through Tradelink AI	Buyer	Type a request such as "order 2 bags of rice"	The assistant finds products with its tools and shows a confirm card; nothing changes until the buyer taps it	No order is placed without a tap; unknown products are not invented	`src/features/ai/tools.test.ts`
FR-023	Add usual items to an empty cart	Buyer	My usual on the empty cart	Usual items are added at their usual quantities	The cart shows them grouped by supplier	No automated test
FR-024	Browse and filter products	Buyer	Open Products	Search, category and supplier filters; quantity to add to cart	"No products match your search and filters." when empty	`src/features/products/availability.test.ts`, `src/features/products/cart.test.ts`
FR-025	Review the cart before placing	Buyer	Open Cart	Lines grouped by supplier with subtotals, cut-off notes, a "Checked:" line listing anything to fix, total before GST, and the budget after the order	Place is disabled while any line needs fixing	`src/features/ordering/cart-groups.test.ts`, `src/features/ordering/line-view.test.ts`

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-026	Warn about stock and offer a substitute	Buyer	A line asks for more than is available	The line is marked and offers a one-tap fix or "Use {product} instead"	The order cannot be placed for more than is available	`src/features/ordering/line-ops.test.ts`, `src/domain/reader/rules.test.ts`
FR-027	Warn about a price rise	Buyer	A line's price is more than 2% above what the buyer last paid	The line shows "You last paid S$x"	No warning at 2% or less	`src/domain/review.test.ts`, `src/features/ordering/line-view.test.ts`
FR-028	Warn when below a supplier's minimum	Buyer	A supplier's subtotal is under its minimum order	"Below {supplier}'s S$X minimum by S$Y." with usual items that would cover the gap; placing is still allowed	The order is flagged "Below minimum" for both sides	`src/domain/minimum.test.ts`, `src/features/supplier/minimum.test.ts`
FR-029	Show cut-off and earliest delivery	Buyer	A supplier group in the cart	"Order by 16:00 for tomorrow" or "Past cut-off, earliest delivery is {day}", from cut-off, lead time and delivery days in Singapore time	The text changes when the cut-off passes	`src/domain/cutoff.test.ts`
FR-030	Place orders and reserve stock	Buyer	Place N orders	One order per supplier is created, stock is reserved for active products, suppliers are notified, the cart is cleared	A shortfall stops everything and highlights the short lines; success shows "N orders placed and stock reserved."	`tests/concurrency/reservations.test.ts`, `e2e/order-flow.spec.ts`
FR-031	Never place the same order twice	Buyer	Placing again with the same idempotency key (double tap, retry)	The stored result is returned and no second order is created	One order per supplier per placement	`tests/concurrency/reservations.test.ts` (stock), no test for the key itself
FR-032	Add a missing product	Buyer	Add to catalogue in the cart, or the Tradelink AI form card	An unconfirmed product is created for a connected supplier with stock 0; the supplier is notified ("Product added")	The line waits for the supplier to confirm; a duplicate name is refused	No automated test
FR-033	Add a supplier not on Tradelink	Buyer	The Tradelink AI "new supplier" form card	An invited business and connection are created; products can then be added for it	No email or message is sent to the supplier; admins see a follow-up after 24 hours	`src/features/ai/tools.test.ts`

6.4 Buyer: orders and after delivery

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-040	List and filter orders	Buyer	Open Orders	Filters All, Active, To check, Delivered, Problems, Cancelled and rejected; search by order number or supplier	Banners show credits to confirm and problems waiting	`src/features/buyer/orders/filters.test.ts`, `src/features/buyer/orders/summary.test.ts`
FR-041	Show an order's progress	Buyer	Open an order	Progress (Placed, Accepted, Out for delivery, Delivered), items, delivery dates, the rest of a two-part delivery, problems and the conversation	Every status has a word, not only a colour	`e2e/pages.spec.ts`
FR-042	Cancel a pending order	Buyer	Cancel order	The order is cancelled, held stock is released, the supplier is notified	Not possible once accepted	No automated test
FR-043	Confirm All here	Buyer	All here on a delivered or out-for-delivery order	The order is marked delivered if needed, then checked; usual "in shop" amounts go up	The order leaves "To check"	No automated test
FR-044	Report Not here yet	Buyer	Not here yet on an order out for delivery	A late problem is opened and the supplier is notified	A second open late report shows "already reported"	No automated test
FR-045	Report a problem on a delivered line	Buyer	Report a problem (type, quantity, note)	A problem opens with amount = quantity × unit price; the supplier is notified	Types: short, damaged, wrong item, expired; the note is at most 500 characters	`e2e/order-flow.spec.ts`
FR-046	Confirm a credit arrived	Buyer	Mark credit received on a problem with a promised credit	The problem moves to credit received; the supplier is told	A promised credit stays open until the buyer confirms it	`e2e/order-flow.spec.ts`, `src/features/buyer/orders/summary.test.ts`, `src/features/notifications/routes.test.ts`
FR-047	Handle a part-accepted order	Buyer	The supplier sends less	The order page says what the supplier cannot send; with a later date, "The rest comes {date}"	The buyer can cancel the rest while it is waiting	`src/domain/todos.test.ts`

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-048	Ask to move a delivery date	Buyer, supplier	Request a new date on an order (first delivery or the rest)	The other side gets a request with Accept and Decline; a new request replaces an open one for the same part	Only the other side can answer; accepting moves the date	`src/features/ai/tools.test.ts`
FR-049	Send an order message	Buyer, supplier	Send a message on an order	The other side is notified with a Reply button	At most 30 messages per person per hour	No automated test

6.5 Buyer: insights and settings

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-050	Summarise spending	Buyer	Open Insights	An "In short" line, figures (spent this week, last week, budget, left, overcharges caught, price changes), a 5-week spend chart, spend by supplier, on-time record, price changes	Rejected and cancelled orders never count as spend	`src/domain/insights.test.ts`, `src/features/insights/summary.test.ts`
FR-051	Track the weekly budget	Buyer	Home and Cart	Spend so far, the pace to Sunday and the budget after the order	Good below 75% used, warning from 75%, over budget above 100%	`src/domain/insights.test.ts`, `src/domain/home.test.ts`
FR-052	Edit budget and alerts	Buyer	Settings > Budget and alerts	Weekly budget and three switches: suggest what to reorder, tell me about price changes, warn me before I go over budget	Budget must be above 0	No automated test
FR-053	Edit usual items	Buyer	Settings > Usual items	Usual order quantity, amount in the shop and use per day per product; remove	Days left update from these numbers	`src/features/buyer/usual-days.test.ts`
FR-054	Acknowledge a price change	Buyer	A price change to-do on Home	The to-do is cleared for this business	It does not come back	`src/domain/todos.test.ts`

6.6 Supplier

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-060	Show what needs the supplier on Home	Supplier	Open Home	Next up: accept new orders, pack today's orders (one pick list added up across orders), answer problems, receive stock, quiet customers; a cut-off countdown and today against a usual weekday	Each step links to the page that does it	`src/domain/home.test.ts`
FR-061	List incoming orders	Supplier	Open Orders	Tabs New, To pack, Out for delivery, Delivered, All; tiles; search	"Below minimum" pill on orders under the supplier's minimum	`src/features/supplier/orderFilters.test.ts`
FR-062	Accept an order	Supplier	Accept order with a delivery date	Stock is reserved for any lines not held yet, those lines are repriced at today's price, the order becomes accepted, the buyer is notified	The date must be today or later; products a buyer added must be confirmed first; a shortfall is refused	`src/features/supplier/dates.test.ts`, `e2e/order-flow.spec.ts`
FR-063	Part accept an order	Supplier	Part accept with a quantity per line	Unsent stock is released. Without a later date, the missing amount is dropped and the buyer gets a to-do. With a later date, the rest is kept as a second delivery (not reserved).	Sending nothing is a rejection, or an error if a later date was given	No automated test
FR-064	Reject an order	Supplier	Reject order	Held stock is released and the buyer is notified	Only while pending	No automated test
FR-065	Deliver an order	Supplier	Mark out for delivery, then Mark delivered	Stock leaves on hand and reserved together; the buyer is notified each time	Works for a suspended supplier finishing accepted orders	`e2e/order-flow.spec.ts`
FR-066	Deliver the rest	Supplier	Mark the rest out for delivery, then Deliver the rest	The rest is taken from free stock and added to the delivered quantities; the buyer checks again	Refused with "not in stock" when free stock is short	No automated test
FR-067	Answer a problem	Supplier	Choose Credit, Replace next delivery, It is on the way (late only) or Dispute	The problem moves to the matching status and the buyer is notified; a dispute also alerts admins	Credit needs an amount above 0; Replace is not offered for late problems	`src/features/supplier/issues.test.ts`, `e2e/order-flow.spec.ts`

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-068	Add a product	Supplier	Add product (name, unit, price, opening stock, reorder level, category)	The product is active with its opening stock recorded as received	Price above 0 and at most 99,999; duplicate names refused	`src/features/supplier/product.test.ts`, `src/features/supplier/validation.test.ts`
FR-069	Receive stock	Supplier	Receive stock with a quantity	On hand goes up and a received movement is recorded; an unconfirmed product becomes active	Quantity must be positive	`src/features/supplier/stock.test.ts`
FR-070	Confirm a product a buyer added	Supplier	Confirm stock (quantity, price, reorder level)	The product becomes active and can be accepted on orders	Only for unconfirmed products	`src/features/supplier/stock.test.ts`
FR-071	Change a price	Supplier	Change price	The new price applies to new orders; a price change is recorded when it differs	Connected buyers with the item as usual see a to-do	`src/features/supplier/validation.test.ts`
FR-072	Adjust stock	Supplier	Adjust stock with a change, reason (damaged, expired, count correction) and note	On hand changes and an adjusted movement is recorded	Refused if on hand would fall below reserved; change between -9,999 and 9,999, not 0	`src/features/supplier/stock.test.ts`
FR-073	See stock history	Supplier	Open Stock history	The latest 200 movements, filterable by type	Every stock change appears	No automated test
FR-074	See activity	Supplier	Open Activity	The latest 200 audit entries about the business, by day, filterable (order, stock, account)	Newest first	`src/features/supplier/activity.test.ts`
FR-075	Edit business profile and terms	Supplier	Settings > Business profile	Contact details, product range, delivery days, lead time (0 to 14 days), cut-off and minimum order are saved	Buyers' cut-off and minimum messages follow the new values	`src/features/supplier/profile.test.ts`

6.7 Admin

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-080	Show the attention queue	Admin	Open Attention	Items derived from 8 rules (section 8), high priority first then oldest; filters All, High, Medium, and Unassigned for the lead	The lead sees all items; an operator sees only items assigned to them	`src/features/admin/attention.test.ts`
FR-081	Show the newest urgent items on Home	Admin	Open Home	Next up shows 4 items, high priority and newest first, and "See all {n} in Attention"	The link opens the Attention page	No automated test
FR-082	Assign an item	Admin lead	Choose an operator in Assigned to	The item is assigned and stays shown after saving	Operators see "Assigned to you" instead of the choice; an operator calling the function is refused	`src/features/admin/attention.test.ts`
FR-083	Mark an item handled	Admin	Mark handled (blocked-action items)	The item is hidden until its cause happens again	Operators can only handle items assigned to them; other items clear when their cause is fixed	`src/features/admin/attention.test.ts`
FR-084	Resolve a dispute	Admin	Credit buyer or Close, no credit	Credit moves the problem to credit promised; close closes it; both sides are notified	Only for disputed problems	No automated test
FR-085	Suspend or reactivate a supplier	Admin	Suspend (with confirmation) or Reactivate	Suspending cancels pending orders, releases their stock, hides the supplier's products from new orders and notifies the supplier and its buyers	A suspended supplier can still deliver accepted orders	No automated test

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-086	View orders, suppliers, buyers, users and operators	Admin	Open each page	Read-only lists with filters and search; operator presence (Online within 5 minutes, Away within 30, else Offline)	Operator workload counts appear only for the lead	`src/features/admin/orderStats.test.ts`, `src/features/admin/presence.test.ts`, `src/features/admin/params.test.ts`
FR-087	Read the audit log	Admin	Open Audit log	30 entries per page, filter by organisation (who made the change) and outcome (Allowed, Blocked)	Entries cannot be edited or deleted	`src/features/admin/params.test.ts`

6.8 Tradelink AI

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-090	Answer from the person's own data	Buyer, supplier	Ask a question	The assistant uses read tools that run as the signed-in person, so it sees only what they can see	Facts come only from tool results; unknown facts are said to be unknown	`src/features/ai/tools.test.ts`
FR-091	Change data only after a tap	Buyer, supplier	A request that changes something	A confirm card describes the change; the change runs only when the person taps it, and is checked again on the server	No write happens from the model alone	`src/features/ai/tools.test.ts`
FR-092	Show working cards	Buyer, supplier	Answers about orders, products, stock, problems or spend	Cards with their own buttons appear in the reply, at most 5	Tags that tools did not return are not drawn	`src/features/ai/ui/reply-blocks.test.ts`
FR-093	Stay available and safe under failure	Buyer, supplier	Claude is not configured, slow or failing	"Tradelink AI is not set up yet." or "Tradelink AI is not available right now…"	No partial changes; the rest of the app keeps working	No automated test
FR-094	Limit use	Buyer, supplier	More than 60 messages in an hour	The request is refused with a rate-limit message	Counted per person	No automated test
FR-095	Keep conversations	Buyer, supplier	Reopen Tradelink AI	Past conversations are listed and reopen scrolled to the latest message	Stored in this browser only	`src/features/ask/history.test.ts`

ID	Requirement	Role	Trigger	Expected behaviour	Acceptance criteria	Test reference
FR-096	Use the assistant on an order page	Buyer, supplier	The chat panel on an order	The assistant gets that order's facts, including why it is flagged (for example below minimum)	Answers refer to that order	No automated test

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7. User journeys

J-01 Ordering (buyer)

Step	Actor	Action	System response	Exception
1	Buyer	Opens Home	Shows "Place tomorrow's order" when usual items run low	No usual items: no plan; the buyer uses Tradelink AI or Products
2	Buyer	Taps Review in cart, types to Tradelink AI, or adds from Products	Lines are added to the cart; Tradelink AI shows a confirm card first	Tradelink AI unavailable: the buyer uses Products and the cart
3	Buyer	Opens the cart	Groups by supplier; shows the Checked line, cut-offs, below-minimum warnings, budget after order	Lines over stock or unmatched block placing until fixed
4	Buyer	Taps Place N orders	Locks stock, creates one order per supplier, reserves stock, notifies suppliers	Another buyer took the stock: STOCK_SHORT, short lines highlighted, nothing placed
5	Supplier	Receives "New order"	Order appears under New	—

J-02 Supplier fulfilment

Step	Actor	Action	System response	Exception
1	Supplier	Opens a new order	Shows lines, stock, the below-minimum flag and products a buyer added	—
2a	Supplier	Accept order with a date	Reserves any unheld lines, reprices them, status Accepted, buyer notified	Date in the past refused; unconfirmed products must be confirmed first; stock short refused
2b	Supplier	Part accept, now only	Releases unsent stock, removes the missing amount, buyer gets a to-do	Sending nothing equals a rejection
2c	Supplier	Part accept with a later date	Keeps the rest as a second delivery (not reserved)	Sending nothing now with a later date is refused
2d	Supplier	Reject	Releases stock, buyer notified	Only while pending

Step	Actor	Action	System response	Exception
3	Supplier	Mark out for delivery, then Mark delivered	Buyer notified each time; stock leaves on hand and reserved	—
4	Supplier	Mark the rest out for delivery, then Deliver the rest	Rest taken from free stock; buyer checks again	Free stock short: "not in stock"; see BUG-01 in section 18
5	Either	Ask to move a date or send a message	Other side notified with Answer or Reply	Only the other side can answer a date request

J-03 Delivery problems and credits

Step	Actor	Action	System response	Exception
1	Buyer	Taps All here	Order checked; done	—
1b	Buyer	Taps Not here yet (out for delivery)	Late problem opened; supplier notified	Already reported: refused
2	Buyer	Reports a problem on a line	Problem opened with amount; supplier notified	Order out for delivery: marked delivered first
3	Supplier	Answers: credit, replace, on the way or dispute	Status changes; buyer notified; dispute alerts admins	Credit needs an amount above 0
4	Buyer	Taps Confirm credit when the credit arrives	Status credit received; supplier told	Until then the problem stays open as "Confirm credit"

J-04 Invoice checks — Not built

Step	Actor	Action	System response	Exception
—	—	—	There is no invoice upload, reading or check. The `invoice` problem type cannot be created.	See section 18

J-05 Adding a supplier who is not on Tradelink (the built part of supplier invitations)

Step	Actor	Action	System response	Exception
1	Buyer	Asks Tradelink AI to add a supplier, or a product from a new supplier	A form card appears	—
2	Buyer	Taps the card's button	An invited business, empty supplier profile and invited connection are created; reply says Tradelink will invite them	Name already connected or listed: refused as a duplicate
3	Buyer	Adds products for that supplier and orders them	Cart lines note "Waits for the supplier to accept your invitation"	—
4	System	After 24 hours	An admin attention item "{supplier} has not joined"	—

Step	Actor	Action	System response	Exception
5	Admin lead	Assigns the item	The operator sees it	No action can contact or activate the supplier
6	Supplier	Signs up under the same business name	Refused as a duplicate	Dead end; see BUG-02 in section 18

J-06 Disputes

Step	Actor	Action	System response	Exception
1	Supplier	Answers a problem with Dispute	Status Disputed; buyer sees "Under review"; admins get "Dispute opened"	—
2	System	—	High-priority attention item "Dispute on ORD-x" with both businesses, type, amount and note	—
3	Admin lead	Assigns it, or handles it	Assigned operator sees it	—
4	Admin	Credit buyer, or Close, no credit	Credit promised or Closed; both sides notified	Only for disputed problems
5	Buyer	Confirms the credit when it arrives (if credited)	Credit received	—

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8. Business rules

ID	Rule	Applies to	Reason
BR-001	Money is stored as integer cents in the app and `numeric(12,2)` in the database; amounts show as S$ with two decimals	All amounts	Avoid rounding errors (ADR 0019)
BR-002	Dates, cut-offs and weeks use Singapore time (Asia/Singapore); weeks run Monday to Sunday	All dates	All businesses are in Singapore (ADR 0019)
BR-003	One person belongs to one business; the business kind decides the role	Accounts	Simple tenancy (ADR 0005)
BR-004	A new account is connected to every active business of the other kind	Sign-up	A new account can trade at once (ADR 0024)
BR-005	Business names are unique per kind, including invited businesses	Sign-up, adding a supplier	Prevent duplicates
BR-006	Passwords need at least 12 characters	Sign-up, reset	ADR 0011
BR-007	Buyers see products only from suppliers they trade with (plus products on their own orders)	Products	Tenancy (ADR 0005)
BR-008	A cart holds at most 60 lines	Cart	Keep orders reviewable

ID	Rule	Applies to	Reason
BR-009	Placing creates one order per supplier	Ordering	Each supplier handles its own order
BR-010	Available stock = on hand − reserved; an order cannot take more than is available	Ordering, accepting	No overselling (ADR 0006)
BR-011	Stock for active products is reserved when the order is placed; stock for products a buyer added is reserved when the supplier accepts	Ordering	The supplier has not confirmed unconfirmed stock (ADR 0006)
BR-012	Below a supplier's minimum order the buyer is warned but never blocked; the order is flagged "Below minimum"	Ordering	Decision D5 (ADR 0020)
BR-013	A line priced more than 2% above what the buyer last paid shows a warning	Cart	BUY-17
BR-014	The reorder plan includes usual items with fewer than 3 days left and tops them up to cover 7 days, limited by available stock	Home	DISCOVERY 6.3
BR-015	The weekly budget defaults to S$900; use is "good" below 75%, "warning" from 75% to 100%, "over" above	Budget	—
BR-016	A buyer can cancel only while the order is pending	Orders	After acceptance the supplier has committed stock
BR-017	The supplier's delivery date must be today or later	Accepting	—
BR-018	Products a buyer added must be confirmed by the supplier before the order can be accepted	Accepting	CONFIRM_STOCK_FIRST
BR-019	Accepting reprices lines that were not held at the supplier's current price	Accepting	Unconfirmed products had no agreed price
BR-020	In a part accept, each line sends 0 up to the ordered quantity; sending nothing at all is a rejection	Part accept	—
BR-021	The rest of a two-part delivery is not reserved; it is taken from free stock when delivered	Second delivery	Stock may arrive later
BR-022	Only the side that did not ask can answer a date request; a new request for the same part replaces the open one	Date changes	—
BR-023	The rest can be cancelled only while it is waiting	Second delivery	—
BR-024	Each person can send at most 30 order messages per hour	Messages	Abuse protection (ADR 0014)
BR-025	Problems can be reported only on delivered lines; reporting on an order out for delivery marks it delivered first	Problems	—
BR-026	A problem's amount = quantity × unit price	Problems	—
BR-027	Only one open "late" problem per order	Not here yet	—
BR-028	Credit needs an amount above 0; "Replace next delivery" is not offered for late or invoice problems; "It is on the way" is only for late problems	Answers	—

ID	Rule	Applies to	Reason
BR-029	A promised credit stays open until the buyer confirms it arrived	Credits	Owner request, 4 November 2026
BR-030	Only admins resolve disputes, by credit or close	Disputes	—
BR-031	Stock can only be adjusted with a reason, by −9,999 to 9,999 and not 0, and never below reserved	Inventory	BELOW_RESERVED (ADR 0006)
BR-032	Prices are above 0 and at most 99,999; a price change is recorded only when the price differs	Inventory	—
BR-033	Suspending a supplier cancels its pending orders, releases their stock and hides it from new orders; accepted orders can still be delivered	Admin	Buyers are not left waiting on a suspended supplier
BR-034	Attention items come from 8 rules: supplier invited more than 24 hours ago (medium), dispute (high), problem open more than 4 hours (medium), product out of stock (high), order pending more than 30 minutes (medium), buyer-added product not confirmed (medium), product at or below reorder level (medium), blocked action (medium)	Admin	Derived items clear themselves when the cause is fixed (ADR 0013)
BR-035	The operations lead sees and assigns every attention item; other operators see only items assigned to them	Admin	Owner request, 4 November 2026
BR-036	"Mark handled" hides an item until its cause occurs again; operators can only handle their own items	Admin	—
BR-037	Tradelink AI changes data only after the person taps a confirm card or a card button, and the change runs as that person	AI	AI must not act on its own (ADR 0008, ADR 0010)
BR-038	Tradelink AI allows 60 messages per person per hour	AI	Cost and abuse control
BR-039	The audit log, stock history, price changes and order events cannot be changed or deleted	Records	Trustworthy history (ADR 0006)
BR-040	Every important action writes an audit entry; refused actions are logged as "denied"	Records	SH-11
BR-041	Critical actions carry an idempotency key, so a retry returns the first result	Writes	OPS-03 (ADR 0004)
BR-042	Order numbers start at ORD-1044 and problem numbers at ISS-302	Records	Continue from the prototype's numbering (Inferred rationale)

9. State models

9.1 Orders

Current state	Event	Next state	Allowed by	Side effects
(none)	Place orders	Pending	Buyer	Stock reserved for active products; supplier notified "New order"

Current state	Event	Next state	Allowed by	Side effects
Pending	Accept or part accept	Accepted	Supplier	Unheld lines reserved and repriced; buyer notified
Pending	Reject (or part accept sending nothing)	Rejected	Supplier	Held stock released; buyer notified
Pending	Cancel order	Cancelled	Buyer	Held stock released; supplier notified
Pending	Supplier suspended	Cancelled	Admin	Held stock released; both sides notified
Accepted	Mark out for delivery	Out for delivery	Supplier	Buyer notified
Out for delivery	Mark delivered, All here, or a problem report	Delivered	Supplier or buyer	Stock leaves on hand and reserved; buyer notified on Mark delivered
Delivered, Rejected, Cancelled	—	—	—	End states

Delivery check (on delivered orders): Unchecked → All here (buyer) or Problem (buyer reports). Delivering the rest sets it back to Unchecked.

9.2 Second delivery ("the rest")

Current state	Event	Next state	Allowed by	Side effects
(none)	Part accept with a later date	Waiting	Supplier	Rest quantities recorded, not reserved; buyer notified
Waiting	Mark the rest out for delivery	Out for delivery	Supplier	Buyer notified
Waiting or Out for delivery	Deliver the rest	Delivered	Supplier	Stock taken from free stock; quantities merged; delivery check reset
Waiting	Cancel the rest	Cancelled	Buyer or supplier	Open date requests for the rest withdrawn; other side notified

9.3 Problems (issues)

Current state	Event	Next state	Allowed by	Side effects
(none)	Report a problem or Not here yet	Open	Buyer	Supplier notified
Open	Credit	Credit promised	Supplier	Buyer notified with Confirm credit
Open	Replace next delivery	Replacement promised	Supplier	Buyer notified
Open (late only)	It is on the way	Supplier replied	Supplier	Buyer notified
Open	Dispute	Disputed	Supplier	Buyer notified; admins get "Dispute opened"

Current state	Event	Next state	Allowed by	Side effects
Disputed	Credit buyer	Credit promised	Admin	Both sides notified
Disputed	Close, no credit	Closed	Admin	Both sides notified
Credit promised	Mark credit received	Credit received	Buyer	Supplier notified
Replacement promised, Supplier replied, Closed, Credit received	—	—	—	End states

9.4 Invoices — Not built

Current state	Event	Next state	Allowed by	Side effects
—	—	—	—	No invoice entity exists. See section 18.

9.5 Invitations (invited businesses)

Current state	Event	Next state	Allowed by	Side effects
(none)	Buyer adds a supplier not on Tradelink	Invited	Buyer	Invited connection created; audit entry; no message sent
Invited	24 hours pass	Invited	System	Admin attention item
Invited	Accept or decline	Active or Declined	—	**Not built.** The database allows these moves but no function performs them.

9.6 Suppliers

Current state	Event	Next state	Allowed by	Side effects
(none)	Sign-up	Active	Supplier	Connected to every active buyer; admins notified
Active	Suspend	Suspended	Admin	Pending orders cancelled and stock released; products hidden from new orders; supplier and buyers notified
Suspended	Reactivate	Active	Admin	Supplier and buyers notified

9.7 Delivery date requests

Current state	Event	Next state	Allowed by	Side effects
(none)	Ask to move a date	Open	Buyer or supplier	Other side notified ("Date change")
Open	Accept	Accepted	The other side	The delivery date moves; asker notified
Open	Decline	Declined	The other side	Asker notified

Current state	Event	Next state	Allowed by	Side effects
Open	A new request for the same part, or the rest is cancelled	Withdrawn	Either side	—

9.8 Products

Current state	Event	Next state	Allowed by	Side effects
(none)	Supplier adds a product	Active	Supplier	Opening stock recorded
(none)	Buyer adds a product	Unconfirmed	Buyer	Supplier notified "Product added"; stock 0
Unconfirmed	Confirm stock or Receive stock	Active	Supplier	Stock recorded

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10. Screen specification

Screen	Role	Purpose	Main content	Actions	Empty or error state
SCR-01 Sign in (`/sign-in`)	Public	Sign in	Email, password	Sign in; links to Create an account and Forgot password	Wrong details, too many attempts, confirm email first, link expired
SCR-02 Create an account (`/sign-up`)	Public	Self sign-up	Kind, business name, name, phone, email, password, confirm	Create account	Field errors; duplicate business name; "Check your email to confirm your account"
SCR-03 Forgot password (`/forgot-password`)	Public	Request a reset	Email	Send link	Always shows "sent"; expired-link notice
SCR-04 Reset password (`/reset-password`)	Public	Set a new password	New password, confirm	Save	Under 12 characters; expired link
SCR-05 Offline (`/offline`)	All	Shown with no network	Title and message	Try again	—
SCR-10 Buyer Home (`/buyer`)	Buyer	What needs the buyer now	Greeting, ask bar, Next up, This week card, glance tabs (Deliveries, Running low, Prices)	Place orders, Review in cart, Not today, Check delivery, to-do actions	"Nothing needs you right now…"
SCR-11 Tradelink AI (`/buyer/assistant`)	Buyer	Ask and act through the assistant	Conversation list, chat, cards, confirm cards	Send, Confirm, card buttons, suggested follow-ups	"Tradelink AI is not set up yet." / "…not available right now"

Screen	Role	Purpose	Main content	Actions	Empty or error state
SCR-12 Cart (`/buyer/cart`)	Buyer	Review and place	Lines by supplier, Checked line, cut-offs, below-minimum, totals, budget bar	Change quantity, swap, remove, Add to catalogue, Place N orders	"Your cart is empty" with My usual and Browse products
SCR-13 Products (`/buyer/products`)	Buyer	Browse and add	Search, category and supplier filters, product grid	Add to cart	"No products match your search and filters."
SCR-14 Orders (`/buyer/orders`)	Buyer	Track orders	Filter chips, search, order list with pills, banners	Open order	Per-filter empty text
SCR-15 Order (`/buyer/orders/[id]`)	Buyer	One order	Progress, items, dates, the rest, problems, conversation, AI panel	Cancel order, Not here yet, All here, Report a problem, Confirm credit, Cancel the rest, date request, message	Not found
SCR-16 Insights (`/buyer/insights`)	Buyer	Spending	In short, figures, chart, by supplier, on-time, price changes	None (read only)	Empty sections say there is no data yet
SCR-17 Alerts (`/buyer/notifications`)	Buyer	All notifications	Tabs All, Needs action, Updates	Mark all read, action buttons	"No updates yet. Changes to your orders show here."
SCR-18 Buyer settings (`/buyer/settings`)	Buyer	Budget, alerts, usual items, suppliers, account, theme	Cards per section	Save budget and switches, edit usual items, Sign out, theme	"You have no usual items yet…", "No suppliers connected yet."
SCR-20 Supplier Home (`/supplier`)	Supplier	What needs the supplier now	Cut-off countdown, Next up, today against a usual weekday, glance tabs (Customers, Running low, Coming up)	Mark all out for delivery, step links	Per-tab empty text
SCR-21 Tradelink AI (`/supplier/assistant`)	Supplier	Ask and act	As SCR-11 with supplier tools	As SCR-11	As SCR-11
SCR-22 Orders (`/supplier/orders`)	Supplier	Incoming orders	Tabs, tiles, search, list or table	Open order	Per-tab empty text
SCR-23 Order (`/supplier/orders/[id]`)	Supplier	Act on one order	Lines, buyer-added products, problems, the rest, conversation	Accept, Part accept, Reject, Out for delivery, Delivered, rest actions, Confirm stock, answer problems, date request, message	Not found

Screen	Role	Purpose	Main content	Actions	Empty or error state
SCR-24 Inventory (`/supplier/inventory`)	Supplier	Stock and prices	Tiles, filters (All, Low, Out, To confirm), product list	Add product, Receive stock, Confirm stock, Change price, Adjust stock	Per-filter empty text
SCR-25 Stock history (`/supplier/stock-history`)	Supplier	Movements	Type filter, latest 200	None	"No stock changes yet."
SCR-26 Activity (`/supplier/activity`)	Supplier	Audit feed	Filters, grouped by day	None	"No activity yet."
SCR-27 Alerts (`/supplier/notifications`)	Supplier	Notifications	List	Mark all read	"No notifications yet."
SCR-28 Supplier settings (`/supplier/settings`)	Supplier	Business profile, account, theme	Profile form	Save, Sign out, theme	Field errors
SCR-30 Admin Home (`/admin`)	Admin	What needs attention	Greeting, Next up (4 items), See all link, tiles, glance tabs (Activity, Low stock, Team online)	Open the queue	Lead: "Nothing needs attention…"; operator: "Nothing is assigned to you…"
SCR-31 Needs attention (`/admin/attention`)	Admin	Work queue	Filters, rows with priority, Assigned to, actions	Assign (lead), Mark handled, Credit buyer, Close no credit, View supplier, Open audit log	Per-filter empty text; operator: "Nothing is assigned to you right now."
SCR-32 All orders (`/admin/orders`)	Admin	Oversight	Status filters, search, tiles	None (read only)	"No orders yet."
SCR-33 Suppliers (`/admin/suppliers`)	Admin	Supplier list	Status filters, list, side panel on laptop	Open supplier, Suspend, Reactivate	"No suppliers yet."
SCR-34 Supplier profile (`/admin/suppliers/[id]`)	Admin	One supplier	Profile, products, orders	Suspend, Reactivate	Not found

Screen	Role	Purpose	Main content	Actions	Empty or error state
SCR-35 Low stock (`/admin/suppliers/low-stock`)	Admin	Products at or below reorder level	List	None	"All products are above their reorder level."
SCR-36 Users (`/admin/users`)	Admin	Buyers and all accounts	Tabs, search	Open buyer	Empty text
SCR-37 Buyer details (`/admin/users/buyers/[id]`)	Admin	One buyer	Details, stats, orders	None	Not found
SCR-38 Operators (`/admin/operators`)	Admin	Team presence	Presence filter, team, shift, assigned items (lead only)	None	"No operators yet."
SCR-39 Audit log (`/admin/audit`)	Admin	Change history	Organisation and outcome filters, 30 per page	Show entries, Clear filters, page links	"No entries match these filters."
SCR-40 Alerts (`/admin/notifications`)	Admin	Notifications	List	Mark all read	Empty text
SCR-41 Admin settings (`/admin/settings`)	Admin	Account and theme	Account card, Appearance	Sign out, theme	—

Redirects: `/buyer/ask` → SCR-11; `/buyer/availability` → SCR-13; `/buyer/order` → SCR-12 or SCR-11; `/buyer/issues/[id]` and `/supplier/issues/[id]` → the order's Problems section.

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11. Notifications

All notifications are in-app (bell panel and Alerts page). There is no email, SMS or push.

Event	Recipient	Channel	Message (kind)	Action
Orders placed	Supplier	In-app	New order	Review order
Order accepted	Buyer	In-app	Order accepted	Open
Order part accepted	Buyer	In-app	Order part accepted	See changes
Order rejected	Buyer	In-app	Order rejected	Open

Event	Recipient	Channel	Message (kind)	Action
Order cancelled by buyer, or by a supplier suspension	Supplier, buyers	In-app	Order cancelled	Open
Out for delivery (first delivery or the rest)	Buyer	In-app	Out for delivery	Open
Delivered (first delivery or the rest)	Buyer	In-app	Order delivered	Check delivery
The rest cancelled	The other side	In-app	Order changed	Open
Date change asked or answered	The other side	In-app	Date change	Answer (when asked), else View order
Order message	The other side	In-app	Message	Reply
Problem reported	Supplier	In-app	Problem reported	Answer
Not here yet	Supplier	In-app	Delivery late	Open
Problem answered	Buyer	In-app	Problem answered	Confirm credit (credit promised), else View order
Dispute opened	Admins	In-app	Dispute opened	Open
Dispute resolved	Buyer and supplier	In-app	Dispute resolved	Open
Credit received	Supplier	In-app	Credit received	View order
Buyer added a product	Supplier	In-app	Product added	Open (Inventory)
Supplier suspended or reactivated	Supplier	In-app	Account suspended / Account reactivated	Open
Supplier suspended or reactivated	Connected buyers	In-app	Supplier suspended / Supplier reactivated	Open
New account signed up	Admins	In-app	New account	Open

Buyers also see to-dos on Home and in the Alerts "Needs action" tab, built from the data rather than from notifications (price changes, part deliveries, replies, credits to confirm, date requests, unanswered messages).

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12. AI behaviour

Tradelink AI is one assistant, powered by Claude (default model `claude-sonnet-5-5`, set with `ANTHROPIC_MODEL`). It runs on the server, as the signed-in person.

Capability	Input	Output	User control	Failure handling	Limits
Answer questions about the person's business	A typed message and up to 20 earlier turns	One or two short sentences plus working cards	Read only	"Tradelink AI is not available right now..."	60 messages per hour; 6 tool rounds per message; 2,000 characters per message
Order for a buyer (find products, add to cart, order now, place the cart)	A typed request	A confirm card naming items, quantities and total	Nothing changes until the buyer taps Confirm; the change is re-validated on the server	Tool errors are reported back plainly; no partial order	Buyer tools: 11 read, 17 confirm, 2 card-only
Act for a supplier (accept, part accept, reject, deliver, answer problems, stock and price changes, terms)	A typed request	A confirm card	Nothing changes until the supplier taps Confirm	As above	Supplier tools: 7 read, 17 confirm, 1 card-only
Add a product or a supplier	A request	A pre-filled form card	The person edits and taps the card's button; the model cannot call these tools itself	Duplicate names refused	Card-only tools
Talk to the other side about an order	A request to message, move a date or answer a date request	A confirm card with the message	Sent only after the tap; the model must not add details the person did not give	—	30 order messages per hour
Explain an order on its page	The order's facts (status, lines, dates, minimum, problems)	Answers about that order, including why it is flagged	Read only unless a confirm card is tapped	—	—
Suggest follow-ups	—	Up to 2 tap-to-ask suggestions	Tapping sends the suggestion as a message	—	—

Protections:

- Text from the database (names, notes) is treated as data, never as instructions.
- IDs are resolved on the server under the person's permissions.
- Writes always need a human tap.
- Usage (tokens, time and outcome) is recorded in `ai_usage`.

Not built:

- invoice reading;
- negotiation drafts;
- an evaluation set;

- the rule-based reader as a fallback inside the live assistant. It exists in the code but has no caller.

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13. Reports and insights

Metric	Definition	Source	Shown to
Spent this week	Total of the buyer's orders placed since Monday 00:00 Singapore time, excluding rejected and cancelled	`src/domain/insights.ts` (`weeklySpend`, `countsAsSpend`)	Buyer (Home, Insights)
Last week	The same for the previous Monday to Sunday	`weeklySpend`	Buyer
Weekly spend chart	Spend per week for the last 4 weeks and this week	`weeklySpend`	Buyer
Pace to Sunday	Where this week's spend lands by Sunday at the pace so far (at least half a day elapsed)	`src/domain/home.ts` (`weekPace`)	Buyer (Home)
Budget and left	Weekly budget and what is left (never below 0); level good below 75%, warning to 100%, over above	`budgetLeft`, `budgetUse`	Buyer
Spend by supplier	Spend per supplier over the 5 weeks, largest first, smallest folded into "Other"; whole percents add to 100	`spendBySupplier`, `src/features/insights/queries.ts`	Buyer
On-time record	Percent of delivered orders that arrived on or before the confirmed date (Singapore date); good at 95% or more, warning from 85%	`onTimePercent`, `onTimeLevel`	Buyer
Overcharges caught	Sum of problem amounts with credit promised or received	`overchargesCaughtCents`	Buyer
Price changes	Changes on the buyer's usual items, newest first, with counts up and down	`priceChangeSplit`, Insights query	Buyer
Days left	(in shop + on order) ÷ use per day, one decimal	`src/domain/reorder.ts` (`daysLeft`)	Buyer
Today against a usual weekday	Order value and new orders so far today against the average of the same weekday over the last 4 weeks, up to the same time	`src/domain/home.ts` (`usualDay`)	Supplier
Customer pulse	Quiet: twice the buyer's usual gap (and at least 2 days) since their last order. Growing: at least 30% more orders in the last 30 days than the 30 before.	`customerPulse`	Supplier
Pick list	Quantities to pack, added up across today's orders, largest first	`pickList`	Supplier
Open orders and open order value	Orders pending, accepted or out for delivery, and their total	`src/features/admin/orderStats.ts`	Admin, supplier tiles
Operators online	Seen within 5 minutes; Away within 30	`src/features/admin/presence.ts`	Admin

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14. Non-functional requirements

ID	Category	Requirement	Verification
NFR-001	Security	Every table has Row Level Security; signed-in users have no insert, update or delete policies; all writes go through checked database functions	Code review of migrations (no write policies found)
NFR-002	Security	The Anthropic key and Supabase secret key are used only on the server and never sent to the browser	Code review (`server-only` modules)
NFR-003	Security	Secrets live in `.env.local` and Vercel settings, never in the repository	`.gitignore`; repository scan on 4 November 2026
NFR-004	Security	Refused actions are logged as denied in the audit log	Code review (`callRpc`, `log_denied`)
NFR-005	Security	Security headers: nosniff, Referrer-Policy, X-Frame-Options DENY, HSTS, Permissions-Policy	`next.config.ts` (no Content Security Policy, see section 18)
NFR-006	Integrity	Stock is never oversold under concurrent orders	`tests/concurrency/reservations.test.ts` (24 parallel placements; opt-in, not run for this document)
NFR-007	Integrity	Critical actions are idempotent	Code review (`app.idem_get`, `app.idem_put`)
NFR-008	Integrity	Status changes follow the state models in section 9	Database triggers (`app.guard_status`); `e2e/order-flow.spec.ts`
NFR-009	Integrity	History tables are append-only	Database triggers (`app.forbid_change`)
NFR-010	Privacy	Logs redact fields named like password, token, secret, key, authorization, cookie, text or email	`src/lib/logger.ts` (no test)
NFR-011	Usability	Copy is sentence case, with no exclamation marks or emoji; buttons name a verb and an object	Design system rule; code review
NFR-012	Accessibility	Skip link, labelled controls, status shown by word as well as colour, `aria-live` for chat, reduced motion respected	Code review; no automated accessibility test
NFR-013	Responsiveness	Every buyer and supplier page fits a phone without sideways scrolling	`e2e/pages.spec.ts` (mobile project, Pixel 7)
NFR-014	Compatibility	Tested in desktop Chrome and a Pixel 7 phone profile	`playwright.config.ts`; other browsers not verified
NFR-015	Resilience	A failed page shows "This page did not load" with Try again; failed admin saves stay on the page	Error boundaries; checked by hand
NFR-016	Resilience	Loss of network shows the offline page	`e2e/smoke.spec.ts`
NFR-017	Cost control	AI chat 60 per hour per person; order messages 30 per hour	Code review (`hit_rate_limit`)

ID	Category	Requirement	Verification
NFR-018	Performance	No response-time or load targets have been measured	Not verified
NFR-019	Availability	No uptime target or monitoring beyond a liveness endpoint (`/api/health`)	Not verified
NFR-020	Maintainability	Strict TypeScript, no `any`, no `console.log`; lint, type-check and unit tests pass before deploy	`npm run lint`, `npm run typecheck`, `npm test` (run locally; there is no CI)

15. Key product decisions

The first eight decisions are summarised in the table. Each one is then explained below, with the problem, options considered, final choice, reasons, tradeoffs and where it appears.

Decision	Options considered	Choice	Reason	Tradeoff
DEC-01 Responsive PWA, not native apps	Native iOS and Android; a responsive web app with a manifest and service worker	Responsive web app (PWA)	One codebase for phone and laptop; no app store	No install prompt or push today; iOS limits
DEC-02 Next.js and Supabase	Pages Router; separate auth provider; an ORM over a direct connection	Next.js App Router on Vercel; Supabase for Postgres, Auth and RLS	Set by the brief; RLS relies on Supabase's `auth.uid()`	Ties the product to two platforms
DEC-03 Permissions in server checks and RLS	UI checks only; RLS-only writes with triggers; writes from TypeScript	Reads through RLS; writes only through checked database functions	Permissions hold even if the UI is bypassed	Business rules live in SQL
DEC-04 Buyers review AI output before anything is sent	AI places orders directly; AI drafts for review	AI drafts; a person taps to confirm	AI can misread; orders commit money	One extra tap
DEC-05 Stock reserved in database transactions	Calculate on read; optimistic versions; row locks	Row locks in one transaction, in product order	No overselling, consistent history	Contention under heavy load (not measured)
DEC-06 Suppliers joining	Secure invitations; self sign-up	Self sign-up now; invitation acceptance not built	Easy to try	Anyone can create an account
DEC-07 Invoices checked but never paid	Pay through Tradelink; check only; no invoices	Check-only was decided; nothing built; no payments	Never move money	Invoice protection not available
DEC-08 Light and dark mode from the design system	Hand-coded colours; generated tokens	Tokens generated from the design system	One source for both themes	Theme is per browser

DEC-01 Responsive PWA instead of native apps

Aspect	Detail
Problem	Buyers order from phones in kitchens and from laptops in the office. Suppliers work from both too.
Options considered	Native iOS and Android apps; a responsive web app; a PWA now with a Capacitor shell later
Final choice	One responsive web app, with an app manifest, a hand-written service worker and an offline page (ADR 0016, ADR 0026)
Why	One codebase and one release for every device. No app store review. Works from a link. ADR 0016 plans a Capacitor shell later.
Tradeoffs	There is no install prompt and push is not switched on. On iPhone, push needs the app on the Home Screen. Offline use is limited to an offline page.
Where it appears	`app/manifest.ts`, `public/sw.js`, `src/pwa/`; phone bottom tabs and laptop top bar (`src/ui/shell/`)

DEC-02 Next.js and Supabase

Aspect	Detail
Problem	A small team needs a secure web app with a database, sign-in and row-level permissions.
Options considered	Next.js Pages Router; Edge runtime everywhere; a separate auth provider (Clerk, Auth0); Prisma or Drizzle over a direct connection
Final choice	Next.js App Router on Vercel with strict TypeScript (ADR 0002); Supabase for Postgres, Auth and Row Level Security (ADR 0003)
Why	The brief fixes both. Server Components send less JavaScript to phones. One identity source keeps RLS simple, because policies use `auth.uid()`. An ORM over a direct connection would bypass RLS.
Tradeoffs	The product depends on two hosted platforms. Business rules are split between SQL and TypeScript.
Where it appears	`app/`, `src/data/`, `supabase/migrations/`

DEC-03 Permissions enforced by server checks and RLS

Aspect	Detail
Problem	Buyers, suppliers and admins share one database. A hidden button is not a permission.
Options considered	Checks in the UI only; RLS-only writes with triggers; several queries from TypeScript
Final choice	Reads use the person's session and RLS. Writes go only through `security definer` database functions that check the caller, the state change and the inputs, and write the history in the same transaction (ADR 0004, ADR 0005). The request proxy and `requireRole` only redirect; they are a convenience.
Why	A permission holds even if someone calls the API directly. Multi-row changes (orders, stock, audit) commit or fail together.
Tradeoffs	Rules live in SQL, which is harder to unit test. There are no pgTAP tests yet.

Aspect	Detail
Where it appears	Every RPC in `supabase/migrations/`; `src/data/rpc.ts` (`callRpc`); `proxy.ts`

DEC-04 Buyers must review AI output before an order is sent

Aspect	Detail
Problem	AI can misread a message or pick the wrong product, and an order commits the buyer to spend.
Options considered	AI places orders directly; AI drafts and a person confirms
Final choice	Tradelink AI can only propose. Each change appears as a confirm card, and runs only when the person taps it, as that person, with the input checked again on the server (ADR 0008, ADR 0010).
Why	The person stays in control and accountable. A prompt-injection attack cannot place an order on its own.
Tradeoffs	One extra tap. "Order now" and Home's "Place tomorrow's order" confirm with a single tap, without the full cart review (Inferred rationale: speed for routine orders). ADR 0010 says the model "has no tools that write"; as built, the model has write tools, but every one is gated by a confirm tap.
Where it appears	`src/features/ai/registry.ts`, `src/features/ai/execute.ts`; confirm cards in Tradelink AI

DEC-05 Stock reserved in database transactions

Aspect	Detail
Problem	Two buyers may order the last units at the same moment. Stock must never be oversold, and every change must be explained.
Options considered	Calculating balances from history on each read; optimistic versions; row locks
Final choice	`inventory_levels` holds on hand and reserved. Every stock change locks the rows in product order, checks availability again, updates the balances and appends a `stock_movements` row, all in one transaction (ADR 0006).
Why	Row locks are simple and contention is low. The constraint `reserved <= on_hand` is a backstop. The history can be replayed to check the balances.
Tradeoffs	Locks can wait under heavy load (not measured). The rest of a two-part delivery is not reserved.
Where it appears	`place_orders`, `accept_core`, `app.lock_inventory`, `app.move_stock`; `tests/concurrency/reservations.test.ts`

DEC-06 How suppliers join: secure invitations, replaced by self sign-up

Aspect	Detail
Problem	Suppliers must join so their buyers can order from them, and a business should only trade with the businesses it chose.
Options considered	Invitation-only onboarding with single-use, hashed, expiring tokens (ADR 0011); open self sign-up

Aspect	Detail
Final choice	Self sign-up (ADR 0024). A new account is connected to every active business of the other kind. A buyer can add a supplier who is not on Tradelink as an *invited* record. **Invitation acceptance is not built**: no email, no token, no accept or decline page.
Why	The owner chose to make Tradelink easy to try (3 November 2026). Secure invitations remain the intended design before real businesses rely on Tradelink (ADR 0024, Consequences).
Tradeoffs	Anyone can create an account and see connected suppliers' prices. An invited supplier who signs up under the same name is blocked (BUG-02).
Where it appears	`/sign-up`; `app.handle_new_user`; `buyer_add_supplier`; the admin attention rule "has not joined"

DEC-07 Invoices are checked but never paid

Aspect	Detail
Problem	Invoices can be wrong or carry changed bank details (fraud). Paying through the app would make Tradelink handle money.
Options considered	Pay invoices through Tradelink; check invoices only; leave invoices out
Final choice	The recorded decision is check-only, never pay (decision D6; ADR 0021). **Invoice checking is not built.** On 4 November 2026 the owner also decided not to integrate Stripe or any payments.
Why	Never holding or moving money keeps risk and regulation low. Checks alone would catch overcharges and bank-detail changes.
Tradeoffs	Buyers get no invoice protection today. The `invoice` problem type exists in the database but cannot be created.
Where it appears	Nowhere in the live product; see section 18

DEC-08 Light and dark mode use the current design system

Aspect	Detail
Problem	Kitchens and offices have very different light. Colours must stay accessible in both themes and match the design.
Options considered	Copy colours into the Tailwind config by hand; generate tokens from the design system
Final choice	`npm run tokens` generates `src/styles/tokens.css` from `design-handoff/design-system/tokens.json`, with light values, dark values and a System mode that follows the device (ADR 0015). A cookie remembers the choice so the server renders the right theme with no flash.
Why	One source of truth for both themes; a design change is a JSON edit and a rebuild.
Tradeoffs	ADR 0015 planned to store the theme on the profile. As built, it is per browser (cookie); the profile column and `set_theme` function are unused.
Where it appears	Top-bar toggle and Settings > Appearance; `src/ui/ThemeToggle.tsx`, `app/layout.tsx`

Further decisions

Decision	Options considered	Choice	Reason	Tradeoff
DEC-09 One order per supplier	One combined order; one per supplier	One per supplier, placed together	Each supplier accepts, delivers and answers for its own goods	A cart can create several orders
DEC-10 Below-minimum orders are allowed	Block; warn	Warn and flag "Below minimum" (ADR 0020)	Decision D5: buyers sometimes need a small order	The supplier may still reject; negotiation not built
DEC-11 Lead and operators in Needs attention	Everyone sees everything; lead assigns and operators see their own	Lead sees and assigns all; operators see their own	Owner request, 4 November 2026	Inferred rationale: clear ownership of each item. No ADR records it.
DEC-12 No payments or Stripe	Subscription billing; payments; none	None	Owner decision, 4 November 2026	No revenue features
DEC-13 Two deliveries for a part-accepted order	Drop what cannot be sent; keep the rest for a later date	Supplier chooses either	Owner request, 4 November 2026	The rest is not reserved
DEC-14 Notifications in-app only	Email; push; in-app	In-app only for now	Inferred rationale: no email provider is configured and push is not switched on; the sprint scope said "no real integrations"	People see updates only when they open the app

16. Assumptions, dependencies and risks

16.1 Assumptions

ID	Assumption
A-01	All businesses and users are in Singapore and trade in Singapore dollars.
A-02	Each person belongs to one business.
A-03	Demo data is fictional. The demo accounts share one password that is never stored in the repository.
A-04	Supabase email confirmation is a project setting; the app handles it on or off.

16.2 Dependencies

Dependency	Used for	If unavailable
Supabase (Postgres, Auth)	All data, sign-in and permissions	The app cannot work
Vercel	Hosting	The app cannot work
Anthropic Claude API	Tradelink AI	Tradelink AI shows "not available"; the rest of the app works

16.3 Risks

ID	Risk	Impact	Current mitigation
R-01	Open sign-up lets anyone see connected suppliers' prices	Data exposure at the demo stage	Accepted for the demo (ADR 0024); bring back invitations or approval before real use
R-02	No two-factor sign-in for admins	Account takeover	Accepted for now (ADR 0024)
R-03	AI misreads a request	A wrong order or change	Confirm before every write; server-side checks
R-04	No CI; checks run by hand	A broken build reaches production	Lint, type-check and tests run before each push
R-05	Stock double-count when the rest is delivered first (BUG-01)	Wrong stock figures	None yet
R-06	Shared database for local development and production	Test data in live records	Demo data is fictional; tests that write are opt-in
R-07	Pages left open across a deploy call actions that no longer exist	A failed save	Admin saves show "Reload the page"; buyer and supplier pages are not covered

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17. Requirements traceability

Status: **Built**, **Partial**, **Not built**. A requirement of "—" means none is defined because the story is not built.

User story	Requirement	Screen	Test	Status
SH-01 App shell and navigation	FR-010, FR-011	All	`src/features/shell/nav.test.ts`	Built
SH-02 Home page pattern	FR-020, FR-060, FR-081	SCR-10, SCR-20, SCR-30	`src/domain/home.test.ts`	Built
SH-03 Theme	FR-012	All	No automated test	Built (per browser)
SH-04 Responsive layout	FR-018	All	`e2e/pages.spec.ts`	Built
SH-05 Confirmations	FR-085, FR-091	SCR-11, SCR-33	`src/features/ai/tools.test.ts`	Built
SH-06 Loading, empty and error states	FR-015, FR-016	All	`e2e/pages.spec.ts`	Built
SH-07 Offline	FR-017	SCR-05	`e2e/smoke.spec.ts`	Partial
SH-08 Accessibility baseline	NFR-012	All	No automated test	Partial
SH-09 Notifications for every role	FR-013, FR-014	SCR-17, SCR-27, SCR-40	`src/features/notifications/routes.test.ts`	Partial (in-app only)

User story	Requirement	Screen	Test	Status
SH-10 Live updates	—	—	—	Not built
SH-11 Audit logging	FR-087	SCR-39	No automated test	Built
SH-12 Writing and formatting	NFR-011	All	`src/domain/money.test.ts`	Built
SH-13 Install as an app	—	—	`e2e/smoke.spec.ts` (manifest served)	Partial
SH-14 Team members	—	—	—	Not built
AUTH-01 Sign in	FR-001	SCR-01	`e2e/smoke.spec.ts`	Built
AUTH-02 Sign out	FR-002	All	No automated test	Built
AUTH-03 Reset a forgotten password	FR-003, FR-004	SCR-03, SCR-04	`src/features/auth/password.test.ts`	Built
AUTH-04 Two-factor for admins	—	—	—	Not built (removed)
AUTH-05 Optional two-factor	—	—	—	Not built
AUTH-06 Role-based access	FR-006, FR-007	All	`src/features/auth/routing.test.ts`	Built
AUTH-07 Accept an invitation	—	—	—	Not built
AUTH-08 Recent sign-in for sensitive changes	—	—	—	Not built
AUTH-09 Session expiry	FR-006	All	`src/features/auth/routing.test.ts`	Built (Supabase sessions)
AUTH-10 Create an account	FR-005	SCR-02	`src/features/auth/signUp.test.ts`	Built
BUY-01 Buyer home	FR-020	SCR-10	`src/domain/home.test.ts`	Built
BUY-02 To do: price changes	FR-020, FR-054	SCR-10	`src/domain/todos.test.ts`	Built
BUY-03 To do: part deliveries	FR-047	SCR-10, SCR-15	`src/domain/todos.test.ts`	Built
BUY-04 To do: replies to problems	FR-020, FR-046	SCR-10	`src/domain/todos.test.ts`	Built
BUY-05 To do: invoice too high	—	—	—	Not built
BUY-06 To do: do not pay yet	—	—	—	Not built
BUY-07 To do: decisions waiting	FR-020, FR-048	SCR-10	`src/domain/todos.test.ts`	Built
BUY-08 Ready for tomorrow	FR-021	SCR-10	`src/domain/reorder.test.ts`	Built
BUY-09 Arriving and to check	FR-020, FR-040	SCR-10, SCR-14	`src/features/buyer/orders/filters.test.ts`	Built

User story	Requirement	Screen	Test	Status
BUY-10 Type or paste an order	FR-022	SCR-11	`src/features/ai/tools.test.ts`	Partial
BUY-11 Reading falls back safely	FR-093	SCR-11	`src/integrations/claude/read-order.test.ts` (reader not in live UI)	Partial
BUY-12 Usual items	FR-021, FR-023	SCR-10, SCR-12	`src/domain/reorder.test.ts`	Built
BUY-13 Add items manually	FR-024	SCR-13	`src/features/products/cart.test.ts`	Built
BUY-14 Review lines	FR-025	SCR-12	`src/features/ordering/line-view.test.ts`	Built
BUY-15 Quantities and units	FR-025	SCR-12	`src/features/ordering/line-ops.test.ts`	Built
BUY-16 Not enough stock and substitutes	FR-026	SCR-12	`src/features/ordering/line-ops.test.ts`	Built
BUY-17 Price warning	FR-027	SCR-12	`src/domain/review.test.ts`	Built
BUY-18 Add a missing product	FR-032	SCR-12, SCR-11	No automated test	Built
BUY-19 Add a product from a new supplier and invite them	FR-033	SCR-11	`src/features/ai/tools.test.ts`	Partial (no invitation)
BUY-20 Order summary, cut-offs and budget	FR-025, FR-029, FR-051	SCR-12	`src/domain/cutoff.test.ts`	Built
BUY-21 Delivery dates	FR-048	SCR-15	`src/features/ai/tools.test.ts`	Built
BUY-22 Confirm and reserve stock	FR-030, FR-031	SCR-12	`tests/concurrency/reservations.test.ts`	Built
BUY-23 CartBar	FR-025	SCR-12	`src/features/ordering/cart-groups.test.ts`	Built
BUY-24 Below a supplier's minimum	FR-028	SCR-12, SCR-15, SCR-23	`src/domain/minimum.test.ts`	Built
BUY-25 Negotiation replies and fees	—	—	—	Not built
BUY-26 Price confirmed higher than estimate	FR-062 (repricing on accept)	SCR-15	No automated test	Partial (no buyer prompt)
BUY-27 Orders and invoices page	FR-040, FR-041	SCR-14, SCR-15	`src/features/buyer/orders/filters.test.ts`	Partial (no invoices)
BUY-28 Cancel a pending order	FR-042	SCR-15	No automated test	Built
BUY-29 All here	FR-043	SCR-15	No automated test	Built

User story	Requirement	Screen	Test	Status
BUY-30 Report a problem	FR-045	SCR-15	`e2e/order-flow.spec.ts`	Built
BUY-31 Not here yet	FR-044	SCR-15	No automated test	Built
BUY-32 to BUY-36 Invoices	—	—	—	Not built
BUY-37 Credits	FR-046	SCR-14, SCR-15	`e2e/order-flow.spec.ts`	Built
BUY-38 Insights summary	FR-050	SCR-16	`src/features/insights/summary.test.ts`	Built
BUY-39 Weekly spend chart	FR-050	SCR-16	`src/domain/insights.test.ts`	Built
BUY-40 Spend by supplier and record	FR-050	SCR-16	`src/domain/insights.test.ts`	Built
BUY-41 Price changes feed	FR-050	SCR-16	`src/domain/insights.test.ts`	Built
BUY-42 Availability	FR-024	SCR-13	`src/features/products/availability.test.ts`	Built (as Products)
BUY-43 Weekly budget	FR-051, FR-052	SCR-18	`src/domain/insights.test.ts`	Built
BUY-44 Alerts and suggestions	FR-052	SCR-18	No automated test	Built
BUY-45 What Tradelink can do	FR-090	SCR-11	No automated test	Built
BUY-46 My usual items	FR-053	SCR-18	`src/features/buyer/usual-days.test.ts`	Built
BUY-47 Ask chat	FR-090, FR-095	SCR-11	`src/features/ask/history.test.ts`	Built (as Tradelink AI)
BUY-48 Ask answers questions	FR-090	SCR-11	`src/features/ai/tools.test.ts`	Built
BUY-49 Ask prepares an order	FR-022, FR-091	SCR-11	`src/features/ai/tools.test.ts`	Built
BUY-50 Ask stays within my business	FR-090	SCR-11	`src/features/ai/tools.test.ts`	Built
SUP-01 Supplier home	FR-060	SCR-20	`src/domain/home.test.ts`	Built
SUP-02 Supplier to do	FR-060	SCR-20	`src/domain/home.test.ts`	Built
SUP-03 Incoming orders	FR-061	SCR-22	`src/features/supplier/orderFilters.test.ts`	Built
SUP-04 Accept an order	FR-062	SCR-23	`e2e/order-flow.spec.ts`	Built
SUP-05 Part accept	FR-063	SCR-23	No automated test	Built
SUP-06 Reject	FR-064	SCR-23	No automated test	Built

User story	Requirement	Screen	Test	Status
SUP-07 Out for delivery and delivered	FR-065, FR-066	SCR-23	`e2e/order-flow.spec.ts`	Built
SUP-08 Answer a below-minimum request	—	—	—	Not built
SUP-09 Answer problems	FR-067	SCR-23	`src/features/supplier/issues.test.ts`	Built
SUP-10 Inventory	FR-068 to FR-072	SCR-24	`src/features/supplier/stock.test.ts`	Built
SUP-11 Add a product	FR-068	SCR-24	`src/features/supplier/product.test.ts`	Built
SUP-12 Receive stock	FR-069	SCR-24	`src/features/supplier/stock.test.ts`	Built
SUP-13 Confirm a product a buyer added	FR-070	SCR-24, SCR-23	`src/features/supplier/stock.test.ts`	Built
SUP-14 Change price	FR-071	SCR-24	`src/features/supplier/validation.test.ts`	Built
SUP-15 Adjust stock	FR-072	SCR-24	`src/features/supplier/stock.test.ts`	Built
SUP-16 Stock history	FR-073	SCR-25	No automated test	Built
SUP-17 Activity	FR-074	SCR-26	`src/features/supplier/activity.test.ts`	Built
SUP-18 Business profile	FR-075	SCR-28	`src/features/supplier/profile.test.ts`	Built
SUP-19 Bank details	—	—	—	Not built
SUP-20 Accept an invitation	—	—	—	Not built
SUP-21 Decline an invitation	—	—	—	Not built
SUP-22 Suspended supplier	FR-085	SCR-33	No automated test	Built
ADM-01 Admin home	FR-081	SCR-30	No automated test	Built
ADM-02 Needs attention items	FR-080	SCR-31	`src/features/admin/attention.test.ts`	Built
ADM-03 Filters and tiles	FR-080	SCR-31	`src/features/admin/attention.test.ts`	Built
ADM-04 Assign to an operator	FR-082	SCR-31	`src/features/admin/attention.test.ts`	Built
ADM-05 Mark handled	FR-083	SCR-31	`src/features/admin/attention.test.ts`	Built
ADM-06 Supplier invitation follow-up	FR-080	SCR-31	`src/features/admin/attention.test.ts`	Partial (assign or handle only)

User story	Requirement	Screen	Test	Status
ADM-07 Resolve a dispute	FR-084	SCR-31	No automated test	Built
ADM-08 Admin notifications	FR-013	SCR-40	`src/features/notifications/routes.test.ts`	Built
ADM-09 All orders	FR-086	SCR-32	`src/features/admin/orderStats.test.ts`	Built
ADM-10 Suppliers	FR-086	SCR-33	`src/features/admin/params.test.ts`	Built
ADM-11 Supplier profile	FR-086	SCR-34	No automated test	Built
ADM-12 Suspend or reactivate	FR-085	SCR-33, SCR-34	No automated test	Built
ADM-13 Users	FR-086	SCR-36	No automated test	Built (view only)
ADM-14 Buyer details	FR-086	SCR-37	No automated test	Built (view only)
ADM-15 Invite a buyer business	—	—	—	Not built
ADM-16 Operators	FR-086	SCR-38	`src/features/admin/presence.test.ts`	Built (view only)
ADM-17 System status	—	—	—	Not built
ADM-18 Audit log	FR-087	SCR-39	`src/features/admin/params.test.ts`	Built
ADM-19 Admin settings	FR-012	SCR-41	No automated test	Partial
ADM-20 Connect a buyer and a supplier	—	—	—	Not built
AI-01 Structured, validated order reading	FR-022, FR-091	SCR-11	`src/integrations/claude/read-order.test.ts`	Partial
AI-02 Matching only my catalogue	FR-090	SCR-11	`src/domain/reader/grounding.test.ts`	Partial
AI-03 Quantities, units and clarification	FR-022	SCR-11	`src/domain/reader/grounding.test.ts`	Partial
AI-04 Prompt injection	FR-091	SCR-11	`src/features/ask/respond.test.ts` (older path)	Partial
AI-05 Usage logging	FR-090	—	No automated test	Built
AI-06 Evaluation set	—	—	—	Not built
AI-07 Invoice reading	—	—	—	Not built
AI-08 Negotiation drafts	—	—	—	Not built
OPS-01 Inventory history	FR-073	SCR-25	`tests/concurrency/reservations.test.ts`	Built
OPS-02 No overselling	FR-030	SCR-12	`tests/concurrency/reservations.test.ts`	Built

User story	Requirement	Screen	Test	Status
OPS-03 Idempotent actions	FR-031	—	No automated test	Built
OPS-04 State machines	Section 9	—	`e2e/order-flow.spec.ts`	Built
OPS-05 Health checks	—	—	—	Partial (liveness only)
OPS-06 Structured logs	NFR-010	—	No automated test	Partial
OPS-07 Rate limits	FR-049, FR-094	—	No automated test	Partial
OPS-08 Demo data and accounts	—	—	—	Built
OPS-09 Retention jobs	—	—	—	Not built
OPS-10 Reliable email	—	—	—	Not built
OPS-11 Continuous integration	—	—	—	Not built
OPS-12 Performance	NFR-018	—	—	Not built (not measured)

18. Known limitations and future improvements

18.1 Not built

Area	What is missing	Related stories
Invoices	Upload, AI reading, checks against the order, bank-detail warnings, "do not pay yet", manual entry. The `invoice` problem type cannot be created.	BUY-05, BUY-06, BUY-32 to BUY-36, SUP-19, AI-07
Invitations	Invitation emails, accept and decline pages, claiming an invited business, inviting buyers or colleagues	AUTH-07, SUP-20, SUP-21, ADM-15, SH-14
Two-factor sign-in	Removed for now (ADR 0024)	AUTH-04, AUTH-05, AUTH-08
Notifications	Email and push (push code exists but nothing sends)	SH-09, OPS-10
Negotiation	Below-minimum requests, delivery fees, AI drafts	BUY-25, SUP-08, AI-08
Admin	Approve suppliers, manage users or operators, connect a buyer and a supplier, system status	ADM-15, ADM-17, ADM-20
Live updates	Pages refresh on navigation, not live	SH-10
Change password in Settings	Only "Forgot password" changes a password	—
Install prompt	The browser's own "Add to Home screen" only	SH-13

Area	What is missing	Related stories
Operations	CI, retention jobs, an AI evaluation set, performance budget, a scheduled job runner	OPS-09, OPS-11, OPS-12, AI-06

18.2 Partial behaviour

Area	Limitation
Typed orders	The live assistant reads orders with its tools. The rule-based fallback reader exists in the code but has no caller, so there is no fallback when Claude is unavailable.
Theme	Remembered per browser, not per account.
Rate limits	Only Tradelink AI and order messages are limited.
Stale pages after a deploy	Admin saves show "Reload the page"; buyer and supplier saves still show "This page did not load".
Audit log filter	The Organisation filter shows only changes made by that business, not changes made to it by others.

18.3 Possible defects, recorded for review

These were found during the documentation audit. They have not been fixed; fixing them needs the owner's approval.

ID	Possible defect	Effect	Evidence
BUG-01	`deliver_rest` is allowed while the order is still accepted or out for delivery	If the rest is delivered before the first delivery, stock is counted twice: on hand ends too low and reserved is released too far	Migration `20261104015354` (`deliver_rest`, `deliver_order`)
BUG-02	Sign-up treats an invited business name as taken	An invited supplier cannot join under its own name or take over the invited record	Migration `20261103104222` (`app.handle_new_user`)
BUG-03	Supplier answers for "invoice" problems exist but no flow creates one	Dead path	Migration `20261103085139` (`report_problem` allows four types)
SEC-01	Some later helper functions keep the default execute grant, and later tables have no explicit revoke	Low risk (RLS still blocks writes), but not as documented	Migrations `20261103131531`, `20261103133813`, `20261104015354`
SEC-02	Any signed-in user can call `hit_rate_limit` with any action name	Rows can grow without limit	Migration `20261103085139`
SEC-03	No Content Security Policy header	Weaker protection against injected scripts than ADR 0010 asks for	`next.config.ts`
OPS-01	The job runner endpoint is public and no schedule calls it	Scheduled jobs never run	`src/jobs/http.ts`; `supabase/migrations/pending_ai_job_cron.sql` is not applied

ID	Possible defect	Effect	Evidence
OPS-02	Suspending a supplier and cancelling an order lock rows in opposite orders	A rare deadlock; Postgres would abort one transaction	Migration `20261103085139` (`set_supplier_status`, `cancel_order`)

18.4 Future improvements

Improvement	Why
Bring back secure invitations and admin approval before real businesses use Tradelink	Closes R-01 and BUG-02 (ADR 0024)
Two-factor sign-in for admins	Closes R-02
Invoice checks (without payment)	Delivers DEC-07 as decided
Email or push notifications	People learn about changes without opening the app
CI running lint, type-check, unit and read-only e2e tests on every push	Closes R-04
Reconnect the rule-based reader as a fallback	Ordering keeps working when Claude is unavailable
Move the theme to the profile	Same theme on every device
Update the older documents and ADR statuses	See the Technical Specification appendix "Differences from older documents"

18.5 Differences from older documents

Several documents in `docs/` describe planned designs that differ from the code. Examples:

- invoice, invitation and email tables;
- an `api` schema;
- admin two-factor;
- pgTAP and evaluation suites.

Many ADRs are still marked "Proposed" although they are built, and no ADR records the current tool-using assistant. The full list is in the Technical Specification appendix "Differences from older documents". This document follows the code.

Appendix A. Glossary

Term	Meaning
Buyer	A food business that orders from suppliers
Supplier	A wholesaler that sells to buyers
Admin, operator	A member of the Tradelink operations team
Operations lead	The operator who sees and assigns every attention item

Term	Meaning
Business, organisation	A buyer, supplier or platform company; every person belongs to one
Connection	A trading link between one buyer and one supplier
Invited business	A supplier record a buyer created for a business not yet on Tradelink
Cart, draft	The buyer's lines before placing
Order	One supplier's part of a placement
Pending, accepted, out for delivery, delivered, rejected, cancelled	Order statuses (shown as Placed, Accepted, Out for delivery, Delivered, Rejected, Cancelled)
The rest, second delivery	The part of a part-accepted order that comes later
Delivery check	The buyer's confirmation after delivery: All here, or a problem
Problem, issue	Something wrong with a delivery: short, damaged, wrong item, expired or late
Credit	Money the supplier agrees to take off; promised, then confirmed received by the buyer
Dispute	A supplier's refusal of a problem, resolved by an admin
On hand, reserved, available	Units in stock; units held for open orders; on hand minus reserved
Usual items	Products a buyer orders regularly, with usual quantity, amount in the shop and use per day
Minimum order	A supplier's smallest order value; below it the order is flagged, not blocked
Cut-off	The time by which an order must be placed for the earliest delivery
Needs attention	The admin work queue, derived from rules
Tradelink AI	The assistant inside the app, powered by Claude
Confirm card	A card that shows a proposed change and runs it only when tapped
RLS	Row Level Security: database rules that limit which rows a person can read
Idempotency key	A unique key sent with an action so a retry does not repeat it
PWA	Progressive web app: a website that can work like an app

Appendix B. Items for owner review

Item	Section	What to confirm
Inferred rationale: "Order now" and "Place tomorrow's order" confirm with one tap, without the full cart review	DEC-04	Is a single confirm tap acceptable as "review" for these paths?
Inferred rationale: lead and operator visibility gives clear ownership of each item	DEC-11	The reason, and whether to record an ADR

Item	Section	What to confirm
Inferred rationale: notifications are in-app only because no email provider or push is set up	DEC-14	Whether email or push is planned
Inferred rationale: order numbers start at ORD-1044 and problems at ISS-302 to continue the prototype's numbering	BR-042	The reason
Theme per browser instead of per account	DEC-08	Keep, or move to the profile as ADR 0015 planned
Possible defects BUG-01 to OPS-02	18.3	Which to fix
ADR 0010 says the model has no write tools; the build gates write tools with a confirm tap	DEC-04	Update ADR 0010, or record a new ADR for the current assistant
BUY-26 "Price confirmed higher than my estimate": accepting reprices lines but the buyer is not asked	Section 17	Whether a buyer prompt is needed